

Constraint-Bundle Forcing under Fixed-Domain Evaluability

A Governance Non-Vacuity Theorem under AASC

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May 2026

Abstract

This paper establishes a positive structural theorem following the fixed-domain closure of the null admissible regime. The previous null-regime result showed that total AASC vacuity cannot function as an admissible endpoint, fixed point, external denial point, empty-extension shield, lower generator, mute-nullity shield, skin-only authority, scope exemption, repair target, or same-type counterexample inside admissibility-bearing evaluability. The present paper begins after that closure. It asks what is structurally forced by non-degenerate fixed-domain evaluability.

The answer is non-vacuous invariant constraint architecture. The central antecedent is not an already-governed architecture. It is non-degenerate fixed-domain evaluability: at least one distinction is treated as reuse-relevant across lawful same-domain continuation, comparison, commitment, construction, exclusion, or counterexample burden. From this continuation target the paper derives persistence criteria, identity fixation, transport, quotient discipline, gate authority, domain and predicate identity, boundary typing, standing-bearing eligibility, redescription stability, and fail-closed treatment of standing failure.

“Governance” here means only invariant role-constraint structure sufficient to preserve admissibility-bearing continuation on a fixed domain. No agency, institutionality, intentionality, social regulation, or decision-making is implied. The governance bundle is a typed role-indexed constraint family, not an ontic inventory, concrete occupant, or physical hypothesis. The main theorem proves that any non-degenerate admissibility-bearing continuation regime forces non-vacuous invariant constraint architecture. In particular, an all-rejecting admissibility predicate may have empty object-level extension while still carrying non-vacuous governance content. Empty admission is therefore not empty governance. The result is a constraint-bundle forcing theorem: admissibility-bearing continuation forces role-existence of governance, and the induced governance architecture is unique up to lawful redescription and governance equivalence. It behaves structurally analogously to a representation theorem for admissibility-bearing continuation, but it does not force admitted occupants, concrete instantiation, spacetime, objects, observers, or unrestricted ontology.

Canonical one-line theorem. Any non-degenerate admissibility-bearing continuation regime forces non-vacuous invariant constraint architecture:

$$\text{NonDeg}(D) \rightarrow \text{CB}(G_D) \neq \text{Vac}.$$

Contents

1 Scope and Target Discipline

3

1.1	Licensed readings	3
1.2	Forbidden readings	4
1.3	Fixed-domain posture	4
2	Dependency on the Null-Regime Theorem	4
3	Binding AASC Support Results	5
4	Minimal Antecedent and Target-Class Fixation	6
5	Negative-to-Positive Transition	8
6	Finite All-Rejecting Evaluator Model	9
7	Dependency-Forcing Ladder	10
8	Gate Authority, Domain Fixation, and Extension Priority	12
9	Standing, Transport, Quotient, and Redescription	13
10	Fail-Closed Governance and No Same-Act Repair	14
11	Main Theorem: Governance Non-Vacuity	14
12	Kernel Forcing and Minimality	15
13	ATS Placement: Governance Requires Load-Bearing Content	16
14	Difference from General Structuralism	17
15	AMetric Boundary and No Pre-Governance Promotion	18
16	Governance Equivalence under Redescription	18
17	Fixed-Domain Route Exhaustion	19
18	Objection Matrix	20
19	Closure Ledger	22
20	Canonical Results	23
20.1	Canonical one-line theorem	23
20.2	Canonical structural-forcing formulation	23

20.3 Canonical role-existence formulation	23
20.4 Canonical empty-extension formulation	23
20.5 Canonical uniqueness formulation	23
20.6 Canonical constraint-induction closure formulation	24
20.7 Canonical anti-overclaim formulation	24
21 Short Proof Form	24
22 Paper III Boundary	25
23 Conclusion	25
24 Counterexample-Burden Appendix	25

1 Scope and Target Discipline

This manuscript is the positive structural successor to the null-regime closure theorem. It does not re-prove null closure. It treats the null admissible regime as unavailable and asks what fixed-domain evaluability minimally requires once some distinction is treated as reuse-relevant across lawful same-domain continuation.

The target is deliberately narrow. The theorem concerns role-existence of governance inside a fixed-domain AASC comparison setting. It does not infer concrete ontology from vocabulary. It does not prove that every domain has evaluability in force. It does not prove that any candidate is object-level admitted. It proves only the following conditional:

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} G_D (\text{Gov}_D(G_D) \wedge \text{CB}(G_D) \neq \text{Vac}).$$

Throughout, “governance” means invariant role-constraint structure sufficient to preserve admissibility-bearing continuation on a fixed domain; it does not imply agency, institutionality, intentionality, social regulation, or decision-making. The forced constraint architecture is not introduced as a governing law imposed externally on the domain; it is the invariant preservation structure induced internally by admissibility-bearing continuation itself. The theorem behaves structurally analogously to a representation theorem for admissibility-bearing continuation: continuation-preserving realizations induce invariant-equivalent constraint architecture. The phrase “role-existence” is not ornamental. It is formally indexed to indispensability under preservation of the target phenomenon. A role exists when its removal prevents preservation of the fixed-domain evaluability target, unless an equivalent role is reintroduced under another name.

1.1 Licensed readings

The following readings are licensed:

- (a) a positive structural theorem following null closure;
- (b) a governance non-vacuity theorem for non-degenerate fixed-domain reuse-relevant evaluability;

- (c) a proof that empty admissibility extension does not imply empty governance;
- (d) a role-existence theorem for a typed constraint-bundle;
- (e) a kernel-forcing and kernel-minimality theorem for fixed-domain evaluability;
- (f) a load-bearing-content theorem under ATS decomposition; and
- (g) a bridge from null-exclusion to a later occupant-forcing question.

1.2 Forbidden readings

The following readings are not licensed:

- (a) a proof that physical spacetime is instantiated;
- (b) a proof that a universe, observer, law, object, field, particle, event, or mind exists;
- (c) a proof that $\exists x \text{ Adm}_D(x) = 1$;
- (d) a proof that every formal system must contain an admitted object;
- (e) a proof that all all-rejecting predicates are invalid;
- (f) a proof that role-existence is concrete-existence; or
- (g) a proof of unrestricted ontology.

The positive conclusion is indexed:

$$\exists_{\text{role}} G_D \neq \exists x \text{ Concrete}(x).$$

1.3 Fixed-domain posture

A fixed-domain posture is not an ontological assumption. It is a sameness-of-target discipline. A proposed regime, classifier, or counterexample remains same-scope only if it is offered as operating on the same comparison target rather than changing the target, adding hidden selectors, importing a richer domain, or becoming inert notation. If it changes the comparison target, it is a scope change. If it changes no licensed use, it is bookkeeping. If it carries no target identity or governing authority, it is not a counterexample to the theorem.

This posture is essential because the paper’s target is not raw absence of objects. The target is governance-vacuity inside a regime that treats some distinction as reuse-relevant under fixed-domain continuation. Raw absence may fail to be anything at all without refuting the theorem. A claim, classifier, regime, endpoint, or counterexample that purports to govern reuse-relevant comparison cannot remain raw absence.

2 Dependency on the Null-Regime Theorem

The previous paper established the fixed-domain structural inadmissibility of the null regime. In the notation used here, its canonical result was:

$$\neg \exists N (\text{Adm}_D(N) = 1 \wedge C(N) = \text{Vac}).$$

Equivalently, total AASC vacuity cannot occupy the admitted side of fixed-domain admissibility-bearing evaluability.

This paper imports the following consequences from that closure result:

- N1. **Null endpoint closure.** Total vacuity cannot be the admissible endpoint, fixed point, or closure state of a same-type fixed-domain evaluability regime.
- N2. **Empty-extension closure.** An admissibility-relevant claim that the admissibility extension is empty is either inert notation or an evaluable classification with non-vacuous governance structure.
- N3. **Boundary closure.** Mute nullity remains AMetric pre-admissibility until non-null selector, witness, identity, transport, or gate structure is supplied.
- N4. **Route closure.** External denial, lower generation, skin-only authority, third status, scope exemption, same-act repair, and same-type counterexample routes all either import non-null governance, change domain, lose same-type force, or become inert.

The dependency is not a substitute for the positive proof. Null closure establishes that the null admissible regime is unavailable. The present theorem identifies the positive structure forced by non-degenerate fixed-domain reuse-relevant evaluability:

$$\text{NonDeg}(D) \rightarrow \text{CB}(G_D) \neq \text{Vac}.$$

3 Binding AASC Support Results

The following AASC results are used as inherited support rather than reproved premises. They are stated here in the form needed for this paper.

Source result	Work performed in the present paper
Minimal construction kernel	Non-degenerate construction forces admissibility, standing, reference, and irreversibility; raw sequences are not constructions; same-act repair is impossible; governance equivalence is the correct uniqueness notion.
Constraint architecture	Coherent fixed-domain evaluability forces witness determinacy, re-description structure, transition covariance, eligibility, internal data closure, grounded transport, and kernel instantiation.
Structure of admissibility	The pre-admissible boundary is AMetric; gate existence is forced; the admissibility interface is bivalent; standing is reuse-stable admissibility; the admissible interior is unique up to identity-preserving equivalence.
Standing-admissibility identity	No same-domain classifier genuinely distinct from standing/eligibility survives; admissibility and standing collapse extensionally on the fixed domain.
Bivalence theorem	No third status is available inside the evaluated fragment: every admissibility-relevant candidate is admitted or not admitted.

AMetric boundary theorem	The boundary has no selector, ranking, metric, parameter, canonical representative, licensed operator, or backward transport of interior structure.
ATS decomposition	Admissibility-bearing descriptive content is exhausted by anchor, tensor, and skin; skin is non-authoritative, while authority-bearing governance is anchor or tensor content.
Fixed-domain exhaustion	Same-scope admissibility-bearing variation is exhausted by boundary typing, identity fixation, standing classification, quotient identity, admissible continuation, role decomposition, and fail-closed repair handling.
Boundary-trace fixation	A positive boundary trace is admissible only when declared, stable, bivalent, outcome-independent, and fail-closed; hidden tie-breakers or post-hoc selectors change the domain.
Closure by exhaustion	Same-scope operators cannot totalize, add hidden auxiliary data, or import family-specific selectors without either changing domain, collapsing to bookkeeping, or violating admissibility.

The common lesson of these support results is this: once a regime is fixed as same-domain and reuse-relevant, governance roles are not optional decorations. They are the conditions under which the target phenomenon is preserved.

4 Minimal Antecedent and Target-Class Fixation

This section is deliberately austere. The main theorem must not smuggle its conclusion into its antecedent. The antecedent is not “governed evaluability.” It is non-degenerate treatment of a distinction as reuse-relevant across lawful same-domain continuation.

Definition 4.1 (Fixed-domain comparison setting). A fixed-domain comparison setting D is an intended same-target comparison frame. Items, marks, acts, statuses, or descriptions are treated as belonging to the same comparison locus rather than to unrelated or newly introduced domains. This definition does not itself include an admissibility gate, standing predicate, transport rule, quotient, redescription relation, or fail-closed repair rule.

Definition 4.2 (Reuse-relevant distinction). A distinction δ on D is reuse-relevant iff the regime treats preserving, rejecting, carrying, blocking, or comparing δ as able to matter for later same-domain use. Reuse-relevance is weaker than admission. It says only that the distinction is not merely inert presentation.

Definition 4.3 (In-force fixed-domain evaluability). $\text{InForce}(D)$ holds iff at least one distinction on D is treated as reuse-relevant under fixed-domain comparison. Equivalently:

$$\text{InForce}(D) \quad \text{iff} \quad \exists \delta \text{ Reuse}_D(\delta).$$

This definition does not presuppose Gate_D , Stand_D , Tr_D , Quot_D , Redesc_D , Bound_D , or Fail_D .

Definition 4.4 (Non-degenerate evaluability). $\text{NonDeg}(D)$ holds iff at least one reuse-relevant distinction on D participates in lawful same-domain comparison across admissibility-relevant continuation, commitment, construction, exclusion, or counterexample burden:

$$\text{NonDeg}(D) \quad \text{iff} \quad \exists \delta (\text{Reuse}_D(\delta) \wedge \text{Continue}_D(\delta)).$$

A one-off distinction that is never compared, carried, reused, excluded, or continued may be a thin presentational contrast, but it is not the target of this theorem. The central forcing result is stated for $\text{NonDeg}(D)$, because transport, quotient, and fail-closed treatment are forced by continuation-preservation rather than by idle marking.

Remark 4.5 (No object-level admission is built in). Neither $\text{InForce}(D)$ nor $\text{NonDeg}(D)$ requires $\exists x \in E_D \text{Adm}_D(x) = 1$. An all-rejecting gate may still be in force. The theorem’s target is governance-vacuity, not empty admission.

Definition 4.6 (Target phenomenon). Let P_D denote the target phenomenon: non-degenerate fixed-domain reuse-relevant evaluability on D . To preserve P_D is to preserve the same-domain treatment of at least one distinction as reuse-relevant across lawful continuation without changing the comparison target, reducing the distinction to inert skin, or importing an undeclared richer domain.

Definition 4.7 (Role-existence). A role R exists relative to P_D , written $\exists_{\text{role}} R$, iff removing R prevents preservation of P_D , unless an equivalent role is reintroduced under another name. Formally:

$$\exists_{\text{role}} R \quad \text{iff} \quad \text{Remove}(R) \rightarrow \neg \text{Preserve}_D(P_D),$$

where $\text{Remove}(R)$ means removal without same-use equivalent replacement. Role-existence is indispensability under fixed-domain preservation. It is not concrete-existence.

Definition 4.8 (Structural forcing). For a target phenomenon P_D and structure S ,

$$P_D \Rightarrow_{\text{struct}} S$$

means that every same-domain realization preserving P_D contains an invariant-equivalent of S :

$$P_D \Rightarrow_{\text{struct}} S \quad \text{iff} \quad \forall \mathcal{R} (\text{Preserve}_{\mathcal{R}}(P_D) \rightarrow S \subseteq \mathcal{R}).$$

Here $\subseteq$ means inclusion up to governance equivalence, not literal syntactic inclusion. Thus “forced” in this paper means structural forcing under fixed-domain preservation. It is not set-theoretic forcing, not category-theoretic forcing, not mere semantic entailment, and not metaphysical necessity.

Remark 4.9 (Usage discipline). The terms *forced*, *unique*, *minimal*, and *exhausted* are used only in fixed-domain role senses. “Forced” means structurally indispensable up to invariant equivalence. “Unique” means unique up to governance equivalence. “Minimal” means removal-sensitive under $\text{Preserve}_D(P_D)$. “Exhausted” means factorization through the listed invariant loci, not an empirical enumeration of every possible notation.

Definition 4.10 (Evaluated fragment). The evaluated fragment E_D is the class of candidates, acts, descriptions, regimes, classifiers, or claims on D that are subject to the relevant classification once a gate is forced. A presented item outside E_D that performs no reuse-relevant work is not a counterexample to a theorem about fixed-domain evaluability.

Definition 4.11 (Governance bundle). A governance bundle G_D is a typed role-indexed constraint family by which non-degenerate fixed-domain reuse-relevant evaluability is preserved. The canonical expanded form is:

$$G_D = \langle \text{Gate}_D, \text{Adm}_D, \text{Stand}_D, \text{Id}_D, \text{Tr}_D, \text{Quot}_D, \text{Redesc}_D, \text{Bound}_D, \text{Fail}_D \rangle.$$

The tuple is not a concrete object, primitive algebraic carrier, or ontic inventory. It is a compact representation of forced roles. The tuple notation is representational rather than ontologically committal: it records the invariant-preserving constraints induced by non-degenerate admissibility-bearing continuation.

Symbol	Type	Governance work
Gate_D	gate role	distinguishes licensed from unlicensed same-domain use.
Adm_D	gate predicate	classifies evaluated candidates as admitted or not admitted.
Stand_D	continuation-eligibility relation	fixes eligibility for commitment, continuation, construction, or reuse.
Id_D	identity-fixation relation	preserves the same target under comparison.
Tr_D	lawful transport operator or relation	carries target identity or status through permitted update.
Quot_D	quotient-consistency relation	separates harmless presentation variation from standing-relevant load.
Redesc_D	admissible redescription relation	permits same-scope reformulation without status drift.
Bound_D	boundary-typing constraint	distinguishes pre-admissible, non-evaluated, rejected, and admitted positions.
Fail_D	fail-closed exclusion constraint	blocks same-act repair from standing failure to standing success.

Definition 4.12 (Constraint-bundle content). $\text{CB}(G_D)$ is the admissibility-relevant invariant content by which G_D performs governance work. It is not a list of concrete occupants. It is role-content.

Definition 4.13 (Governance vacuity). A domain is governance-vacuous iff no gate, admissibility predicate, standing role, identity role, transport, quotient, redescription, boundary, or fail-closed role is in force. Notationally:

$$\text{GovernanceVac}_D \quad \text{iff} \quad \text{CB}(G_D) = \text{Vac}.$$

Definition 4.14 (Empty admission and empty governance). Empty admission means that no object-level evaluated candidate is admitted:

$$\forall x \in E_D, \quad \text{Adm}_D(x) = 0.$$

Empty governance means that no role-indexed constraint family is in force:

$$\text{CB}(G_D) = \text{Vac}.$$

The main corollary of this paper is that empty admission does not imply empty governance.

5 Negative-to-Positive Transition

Paper I closed the direct null route:

$$\neg \exists N (\text{Adm}_D(N) = 1 \wedge C(N) = \text{Vac}).$$

The present paper asks what remains when a regime is not null in that sense but may still have empty admission, all-rejection, or no object-level occupants. The positive residue is not a thing, particle, event, observer, or spacetime. It is governance role-content.

The transition is:

$$\text{null admissible closure blocked} \rightsquigarrow \text{governance forced under } \text{NonDeg}(D).$$

It is not:

$$\neg \text{Vac} \rightsquigarrow \exists x \text{ Concrete}(x).$$

The proof strategy is local. We begin with a reuse-relevant distinction under lawful continuation and derive persistence criteria, identity fixation, transport, quotient discipline, gate authority, domain/predicate identity, extension-definability, standing-bearing classification, redescription stability, boundary discipline, and fail-closed failure-handling. Those roles form a typed governance bundle. Since each is indispensable role-content, the bundle is not vacuous.

6 Finite All-Rejecting Evaluator Model

The finite model is included before the general proof because it isolates the central distinction: empty admission is not empty governance.

Definition 6.1 (Two-stage all-rejecting evaluator model). Let M_0 be a finite model with:

$$E_D^0 = \{a_0, b_0\}, \quad E_D^1 = \{a_1, b_1, a'_1\},$$

where u is one intended update from stage 0 to stage 1. Let $=$ distinguish a_0 from b_0 . Let $\tau_u(a_0) = a_1$ and $\tau_u(b_0) = b_1$. Let $a_1 \sim a'_1$ express harmless presentation variation at stage 1. Let Adm_D be the all-rejecting predicate:

$$\text{Adm}_D(a_0) = \text{Adm}_D(b_0) = \text{Adm}_D(a_1) = \text{Adm}_D(b_1) = \text{Adm}_D(a'_1) = 0.$$

Thus:

$$\text{Ext}(\text{Adm}_D) = \emptyset.$$

Nevertheless, the domain, update, transport, quotient, predicate, and rejection/non-evaluation boundary are fixed.

Proposition 6.2 (Empty extension with non-empty governance). *The two-stage all-rejecting evaluator has empty admissibility extension but non-vacuous governance.*

Proof. No evaluated candidate is admitted, so $\text{Ext}(\text{Adm}_D) = \emptyset$. But the model fixes the evaluated fragments, distinguishes evaluated rejection from non-evaluation, identifies the predicate doing the rejecting, transports a_0 to a_1 and b_0 to b_1 , and quotients a_1 with its harmless presentation variant a'_1 . These are governance roles. Hence:

$$\text{Ext}(\text{Adm}_D) = \emptyset \quad \text{and} \quad \text{CB}(G_D) \neq \text{Vac}.$$

The empty object-level extension therefore does not collapse the governance layer. $\square$

Proposition 6.3 (Model-level removal tests). *In M_0 , removing transport, quotient, or gate authority destroys a different structural aspect of the target phenomenon.*

Proof. If transport is removed, there is no fact of the form $\tau_u(a_0) = a_1$. The later occurrence a_1 is not fixed as the same reuse-relevant item as a_0 , so continuation collapses. If quotient consistency is removed, a_1 and a'_1 cannot be identified as harmless presentation variants; presentation drift becomes hidden standing-bearing load. If gate authority is removed, the all-rejecting status is no longer a rejection classification. It becomes undefined status or non-evaluation. Thus even a model with empty admission extension requires non-empty governance. $\square$

Corollary 6.4 (No occupant inference). *The existence of non-vacuous governance does not imply the existence of an admitted object-level candidate. The all-rejecting evaluator is a witness:*

$$\text{CB}(G_D) \neq \text{Vac} \quad \nRightarrow \quad \exists x \in E_D \text{ Adm}_D(x) = 1.$$

Remark 6.5 (Finite-to-general lifting). The finite witness demonstrates the minimal forcing pattern in the smallest non-degenerate evaluability regime: a reuse-relevant distinction under update requires identity, transport, quotient, gate, and boundary discipline even when every evaluated candidate is rejected. The witness is minimal in the sense that removal of any single forcing role destroys continuation-preserving evaluability rather than producing a weaker same-domain evaluator. The finite witness therefore exhibits the minimal continuation-preserving forcing pattern whose invariant structure is generalized by the full theorem.

7 Dependency-Forcing Ladder

The central theorem must not smuggle a full governance bundle into its antecedent. The following ladder records the stepwise structural forcing from non-degenerate reuse-relevance to the expanded governance bundle.

Proposition 7.1 (L1: Reuse relevance forces persistence criteria). *If $\text{NonDeg}(D)$, then the regime must fix criteria by which the reuse-relevant distinction remains the same distinction across admissibility-relevant continuation.*

Proof. $\text{NonDeg}(D)$ says that a distinction is treated as reuse-relevant across lawful same-domain continuation. Without persistence criteria, the later use is not a continuation of the same distinction; it is a new presentation, retargeting, or scope change. Thus persistence criteria are required for the target phenomenon to be preserved. $\square$

Proposition 7.2 (L2: Persistence criteria force identity fixation). *Persistence criteria for a reuse-relevant distinction force identity fixation:*

$$\text{NonDeg}(D) \rightarrow \text{Id}_D.$$

Proof. A persistence criterion must determine what item, status, or distinction is preserved. That determination is identity fixation. Without Id_D , persistence is not persistence of the same target but an untyped recurrence or replacement. $\square$

Proposition 7.3 (L3: Identity fixation under update forces transport). *If identity is to persist across lawful continuation or update, then transport structure is forced:*

$$\text{Id}_D \wedge \text{Continue}_D \rightarrow \text{Tr}_D.$$

Proof. Identity at one stage does not by itself determine how the same target is carried into a later stage. Continuation requires a lawful passage from prior target or status to later target or status. That passage is transport. Without Tr_D , the alleged continuation is an unlicensed re-identification or target loss. $\square$

Proposition 7.4 (L4: Transport under presentation variance forces quotient discipline). *If transported targets can be redescribed or presented in standing-irrelevant variants, then quotient discipline is forced:*

$$\text{Tr}_D \wedge \text{PresentationVariance}_D \rightarrow \text{Quot}_D.$$

Proof. Transport must preserve standing-relevant content without treating harmless presentational variation as new load. A quotient identifies variants that make no licensed-use difference while preserving differences that do. Without quotient discipline, transport either overcounts skin or erases genuine invariant load. $\square$

Proposition 7.5 (L5: Quotient discipline plus continuation forces admissibility gating). *If a regime preserves standing-relevant content through quotient-respecting continuation, then a gate is forced to distinguish licensed from unlicensed standing-bearing use:*

$$\text{Quot}_D \wedge \text{Continue}_D \rightarrow \text{Gate}_D.$$

Proof. Quotient-respecting continuation determines what can be treated as the same standing-relevant content. But continuation still needs a distinction between permitted and blocked uses of that content. Otherwise no candidate is licensed, rejected, continued, or excluded as standing-bearing. That licensed/unlicensed distinction is the admissibility gate. $\square$

Proposition 7.6 (L6: Admissibility gating under continuation forces fail-closed standing treatment). *If a gate licenses standing-bearing continuation, then failed standing must be fail-closed for the same evaluated act:*

$$\text{Gate}_D \wedge \text{Continue}_D \rightarrow \text{Fail}_D.$$

Proof. If a failed standing status could be repaired into positive standing while remaining the same evaluated act, the gate would not determine stable licensed use. Either repair supplies new governance and triggers fresh evaluation, or it changes the act, or it fails. Thus gate-governed continuation requires fail-closed treatment of standing failure. $\square$

Corollary 7.7 (Ladder summary).

$$\text{NonDeg}(D) \Rightarrow_{\text{struct}} \{\text{Id}_D, \text{Tr}_D, \text{Quot}_D, \text{Gate}_D, \text{Adm}_D, \text{Stand}_D, \text{Redesc}_D, \text{Bound}_D, \text{Fail}_D\}.$$

The subsequent sections spell out these roles as the governance bundle G_D .

Proposition 7.8 (Closure of the forcing ladder). *The dependency-forcing ladder is exhaustive for non-degenerate admissibility-bearing continuation. Any same-domain continuation-preserving realization must preserve persistence, identity, transport, quotient discipline, gate authority, and fail-closed standing treatment; each preservation requirement induces the next role in the ladder. Hence no weaker same-domain continuation architecture preserves P_D without reintroducing an invariant-equivalent of the omitted role.*

Proof. Let $\mathcal{R}$ preserve P_D . Since P_D includes reuse-relevant continuation, $\mathcal{R}$ must preserve persistence of the relevant distinction. Persistence requires identity; identity through update requires transport; transport across harmless presentation variance requires quotient discipline; quotient-respecting continuation requires a gate; and gate-governed continuation requires fail-closed handling of standing failure. If any item in this sequence is absent, either the target distinction is no longer preserved, the domain has changed, or an equivalent role is being supplied under another description. These cases exhaust same-domain preservation of P_D . Therefore the ladder is closed. $\square$

8 Gate Authority, Domain Fixation, and Extension Priority

Lemma 8.1 (G1: Non-degenerate evaluability requires gate authority). *If non-degenerate fixed-domain evaluability is in force on D , then a gate role is forced on D :*

$$\text{NonDeg}(D) \rightarrow \text{Gate}_D.$$

Proof. By definition, $\text{NonDeg}(D)$ means that at least one distinction is treated as reuse-relevant across lawful same-domain continuation. By the dependency-forcing ladder, persistence, transport, and quotient-respecting continuation require a licensed/unlicensed distinction. If there is no gate role at all, there is no fixed distinction between permitted and blocked later use, no difference between rejected use and non-evaluation, and no criterion by which the distinction can matter for reuse rather than merely appear as inert presentation. The target phenomenon is thereby lost. Therefore gate authority is forced. $\square$

Lemma 8.2 (G2: Gate authority requires domain and predicate identity). *If a gate role is in force on D , then domain identity, evaluated-fragment identity, predicate identity, and boundary typing are forced:*

$$\text{Gate}_D \rightarrow \text{Id}(D) \wedge \text{Id}(E_D) \wedge \text{PredId}(\text{Adm}_D) \wedge \text{Bound}_D.$$

Proof. A gate cannot classify unless the domain over which it classifies is fixed, the evaluated fragment is distinguished from non-evaluated presentation, and the predicate doing the classification has identity. A gate without domain identity has no field of application. A gate without predicate identity is not the same gate across redescription. A gate without boundary typing cannot distinguish rejection from non-evaluation or admissible interior from pre-admissible proposal. Hence gate authority requires those identity and boundary roles. $\square$

Theorem 8.3 (G3: Governance precedes extension). *The admissibility extension is definable only relative to governance roles fixing a domain, evaluated fragment, predicate identity, and boundary. Thus:*

$$\text{Defined}_D(\text{Ext}(\text{Adm}_D)) \rightarrow \exists_{\text{role}}\{\text{Id}(E_D), \text{PredId}(\text{Adm}_D), \text{Bound}_D\},$$

while governance vacuity blocks extension definition:

$$\text{CB}(G_D) = \text{Vac} \rightarrow \neg \text{Defined}_D(\text{Ext}(\text{Adm}_D)).$$

Proof. The extension $\text{Ext}(\text{Adm}_D)$ is the class of evaluated candidates satisfying the predicate Adm_D . To define it, one must fix the evaluated fragment E_D , identify the predicate Adm_D , and distinguish evaluated rejection from non-evaluation. Those are governance roles. If $\text{CB}(G_D) = \text{Vac}$, none of these roles is in force. Therefore the extension is not defined as an admissibility-relevant object. Governance is prior to extension classification, including empty-extension classification. $\square$

Corollary 8.4 (All-rejection still carries predicate identity). *If $\text{Adm}_D(x) = 0$ for every $x \in E_D$, the classification is still non-vacuous whenever it is admissibility-relevant, because D , E_D , Adm_D , and the rejection/non-evaluation boundary must be fixed before the empty extension is even defined.*

9 Standing, Transport, Quotient, and Redescription

Lemma 9.1 (G4: Standing-bearing classification requires identity stability). *If a classification has standing-bearing force on D , then target identity must be stable under lawful same-scope redescription:*

$$\text{Class}_D(x) \wedge \text{StandingRelevant}_D(x) \rightarrow \text{Id}_D(x) \wedge \text{Redesc}_D(x).$$

Proof. A classification that changes merely because the same target is lawfully redescribed is not a fixed-domain standing-bearing classification. It is presentation-sensitive drift. Standing-bearing force requires that the same reuse-relevant target remain fixed under permitted presentation changes. Thus identity stability and redescription discipline are required. $\square$

Lemma 9.2 (G5: Continuable classification requires transport). *If a classification is available for continuation, commitment, construction, or reuse, then transport discipline is required:*

$$\text{Class}_D(x) \wedge \text{Continue}_D(x) \rightarrow \text{Tr}_D(x).$$

Proof. Continuation requires the same target or same use-relevant status to be carried through a lawful update. Without transport, the later occurrence is not the same standing-bearing item under continuation. It is a reclassification, retargeting, scope change, or identity loss. Therefore transport discipline is forced by continuable classification. $\square$

Lemma 9.3 (G6: Presentation-neutral classification requires quotient consistency). *If fixed-domain evaluability distinguishes standing-relevant from standing-irrelevant presentation change, then quotient consistency is forced:*

$$\text{Redesc}_D \wedge \text{Class}_D \rightarrow \text{Quot}_D.$$

Proof. Presentation-neutral evaluation requires harmless representational variation to be identified while standing-relevant variation remains gated. If harmless presentation differences cannot be quotiented away, skin differences acquire hidden standing-bearing load. If standing-relevant differences are quotiented away, governance erases the very load it must preserve. Either failure destroys fixed-domain evaluability. Therefore quotient consistency is forced. $\square$

Proposition 9.4 (Standing-admissibility identity on the positive side). *On the positive side of a fixed-domain evaluated fragment, admissibility, standing, identity stability, transport consistency, quotient identity, and fixed-domain classificatory legitimacy collapse extensionally:*

$$\text{Adm}_D(x) = 1 \Leftrightarrow \text{Stand}_D(x) = 1 \Leftrightarrow \text{Id}_D(x) = 1 \Leftrightarrow \text{Tr}_D(x) = 1 \Leftrightarrow \text{Quot}_D(x) = 1 \Leftrightarrow \text{Class}_D(x) = 1.$$

Proof. If a candidate is admitted for licensed standing-bearing use but lacks standing, admission fails to license the use it is defined to license. If standing lacks identity, no same target preserves standing. If identity lacks transport, the target cannot be carried through lawful continuation. If transport lacks quotient consistency, presentation differences can become hidden load or genuine load can be erased. Conversely, a candidate with fixed-domain classificatory legitimacy for commitment, continuation, or reuse is admitted with standing-bearing identity, transport, and quotient stability. Thus the positive-side statuses collapse extensionally. $\square$

10 Fail-Closed Governance and No Same-Act Repair

Lemma 10.1 (G7: Standing failure must be fail-closed). *If standing failure occurs within a fixed-domain governance regime, the failure must be fail-closed for the same evaluated act:*

$$\text{Stand}_D(x) = 0 \rightarrow \neg \text{SameActRepair}_D(x).$$

Proof. If an act classified as standing-null can later become standing-positive while remaining the same evaluated act on the same fixed domain, the original classification was not stable. If the later stage supplies a new gate, target, transport rule, quotient identity, anchor, tensor, or invariant bundle, it supplies non-null governance or a fresh evaluation. If it supplies none, no repair occurs. Therefore standing failure is terminal for the same evaluated act. $\square$

Corollary 10.2 (No governance-by-repair). *Governance cannot be generated by repairing a governance-vacuous act while treating it as the same act. Repair either fails, supplies non-vacuous governance, or constitutes fresh evaluation of a new act or scope.*

11 Main Theorem: Governance Non-Vacuity

Theorem 11.1 (G8: Constraint-bundle forcing theorem). *If non-degenerate fixed-domain evaluability is in force on D , then there exists a non-vacuous governance bundle on D :*

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} G_D (\text{Gov}_D(G_D) \wedge \text{CB}(G_D) \neq \text{Vac}).$$

Equivalently:

$$P_D \Rightarrow_{\text{struct}} G_D.$$

Proof. Assume $\text{NonDeg}(D)$. Then some distinction is treated as reuse-relevant across lawful same-domain continuation. By the dependency-forcing ladder, reuse relevance forces persistence criteria, persistence criteria force identity fixation, identity under update forces transport, transport under presentation variance forces quotient discipline, quotient-respecting continuation forces gate authority, and gate-governed continuation forces fail-closed standing treatment. By Lemma G2, gate authority requires domain identity, evaluated-fragment identity, predicate identity, and boundary typing. By Theorem G3, extension classification is definable only relative to those governance roles. By Lemmas G4–G6, standing-bearing classification requires identity stability, lawful redescription discipline, transport, and quotient consistency. By Lemma G7, standing failure requires fail-closed treatment. These roles form a typed governance bundle:

$$G_D = \langle \text{Gate}_D, \text{Adm}_D, \text{Stand}_D, \text{Id}_D, \text{Tr}_D, \text{Quot}_D, \text{Redesc}_D, \text{Bound}_D, \text{Fail}_D \rangle.$$

Each listed component is reuse-relevant invariant constraint content. Removing any component without equivalent replacement destroys, changes, or evacuates P_D . Therefore G_D role-exists and its constraint-bundle content is not vacuous:

$$\text{CB}(G_D) \neq \text{Vac}.$$

Hence:

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} G_D (\text{Gov}_D(G_D) \wedge \text{CB}(G_D) \neq \text{Vac}).$$

$\square$

Corollary 11.2 (No governance-vacuous non-degenerate evaluability). *There is no fixed-domain regime in which non-degenerate evaluability is in force while all governance roles are absent:*

$$\neg(\text{NonDeg}(D) \wedge \text{GovernanceVac}_D).$$

Corollary 11.3 (Empty admission is not empty governance). *For any fixed-domain regime, object-level empty admission does not imply governance vacuity:*

$$\forall x \in E_D \text{ Adm}_D(x) = 0 \quad \not\Rightarrow \quad \text{CB}(G_D) = \text{Vac}.$$

If the empty-extension claim is used as an admissibility-relevant classification, then:

$$\text{Class}_D(\forall x \in E_D \text{ Adm}_D(x) = 0) \rightarrow \text{CB}(G_D) \neq \text{Vac}.$$

Proof. The all-rejecting evaluator model shows consistency of empty object-level extension with non-empty governance. More generally, an admissibility-relevant empty-extension classification fixes the domain, evaluated fragment, predicate, extension notion, rejection/non-evaluation boundary, redescription behavior, and classificatory role. These are governance roles. Therefore the classification is not governance-vacuous. $\square$

Remark 11.4 (The theorem is positive but not ontological). The theorem proves:

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} G_D.$$

It does not prove:

$$\text{NonDeg}(D) \rightarrow \exists x \in E_D \text{ Adm}_D(x) = 1,$$

let alone:

$$\exists x \text{ Concrete}(x).$$

This distinction is the exact boundary of the present theorem.

12 Kernel Forcing and Minimality

Definition 12.1 (Expanded governance kernel). For the present paper, the expanded AASC governance kernel is:

$$K_D = \{\text{Adm}_D, \text{Stand}_D, \text{Id}_D, \text{Tr}_D, \text{Quot}_D, \text{Redesc}_D, \text{Bound}_D, \text{Fail}_D\}.$$

This expanded kernel refines the constructional kernel $\{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$ by spelling out the fixed-domain evaluability roles that carry reference and irreversibility: identity, transport, quotient, redescription, boundary, and fail-closed treatment.

Theorem 12.2 (G9: Kernel forcing). *If non-degenerate fixed-domain evaluability is in force on D , then the expanded governance kernel role-exists:*

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} K_D.$$

Proof. The main theorem forces G_D . The kernel K_D is the subset of G_D consisting of the roles necessary for admissibility classification, standing-bearing continuation, target identity, transport, quotient, redescription, boundary typing, and failure closure. Each role was forced by Lemmas G1–G7 and Theorem G3. Hence K_D role-exists. $\square$

Theorem 12.3 (G10: Kernel minimality). *No role in K_D can be removed while preserving non-degenerate fixed-domain reuse-relevant evaluability:*

$$\forall R \in K_D, \quad \text{NonDeg}(D) \wedge \text{Remove}(R) \rightarrow \text{Collapse}_D.$$

Here Collapse_D means loss of the target phenomenon, scope change, hidden selector import, bookkeeping-only residue, or fresh non-null governance replacement.

The removal map is finite:

Removed role	Structural consequence
Adm_D	No licensed/unlicensed distinction remains at the evaluated interface.
Stand_D	No continuation, commitment, construction, or reuse is standing-bearing.
Id_D	No same target is preserved under comparison.
Tr_D	No lawful update carries target identity or status forward.
Quot_D	Harmless presentation variation either becomes hidden load or genuine load is erased.
Redesc_D	Same-scope reformulation can change status; fixed-domain invariance fails.
Bound_D	Pre-admissible, non-evaluated, rejected, and admitted positions are not typed.
Fail_D	Standing failure can be silently repaired as the same act; classification stability fails.

Proof. The table exhausts the expanded kernel. In each row, removal of the role prevents preservation of P_D unless an equivalent role is reintroduced. Equivalent reintroduction does not defeat minimality; it confirms role-existence under another name. Therefore no proper removal preserves non-degenerate fixed-domain reuse-relevant evaluability. $\square$

Corollary 12.4 (No weaker governance-vacuous kernel). *A proposed weaker package either performs the same governance work up to role-equivalence, changes licensed use and becomes a new gate, changes domain, or fails to support the target phenomenon. It cannot be both same-scope and governance-vacuous.*

13 ATS Placement: Governance Requires Load-Bearing Content

Definition 13.1 (ATS roles). Within a fixed evaluability regime, a descriptive element is:

- (a) an *anchor* when it supplies admissibility-defining preconditions;
- (b) a *tensor* when it supplies quotient-level invariant standing-bearing content; or
- (c) *skin* when it supplies presentation without gate authority.

ATS is exhaustive for admissibility-bearing descriptive content on a fixed domain.

Definition 13.2 (Invariant load). Invariant load is admissibility-relevant structure preserved under lawful same-domain continuation and quotient-respecting redescription. A role carries invariant load

when removing it changes licensed use, standing-bearing continuation, target identity, transport behavior, quotient behavior, boundary status, or fail-closed treatment.

Theorem 13.3 (G11: No skin-only governance). *If a structure governs fixed-domain reuse-relevant evaluability, then it is not skin-only:*

$$\text{Gov}_D(x) \rightarrow \neg \text{SkinOnly}_D(x).$$

Proof. Skin changes presentation without changing licensed use. Governance determines or preserves licensed use: admission, standing, identity, transport, quotient behavior, redescription, boundary typing, or failure handling. A structure cannot both change or preserve licensed use in this authority-bearing sense and be pure presentation surplus. Therefore governance is not skin-only. $\square$

Theorem 13.4 (G12: Governance requires load-bearing content). *Any structure affecting admissibility-bearing continuation contributes anchor or tensor content:*

$$\text{Gov}_D(x) \rightarrow (\text{Anchor}_D(x) \vee \text{Tensor}_D(x)).$$

For a bundle:

$$\text{Gov}_D(G_D) \rightarrow \exists r \in G_D (\text{Anchor}_D(r) \vee \text{Tensor}_D(r)).$$

Proof. Since ATS exhausts admissibility-bearing descriptive roles into anchor, tensor, and skin, and since governance cannot be skin-only by G11, governance-bearing content must be anchor or tensor content. Hence any governance role must contribute either admissibility-defining precondition content, which is anchor content, or invariant standing-bearing content, which is tensor content. No fourth authority-bearing presentation category remains. $\square$

Corollary 13.5 (Blank governance is impossible). *A blank mark, empty label, zero inscription, or merely presentational token cannot itself be the governance bundle. It either has no authority and is skin, or it carries authority through anchor/tensor content and is not governance-vacuous.*

14 Difference from General Structuralism

The theorem is adjacent to structuralist themes, but it is not merely structuralism by relabeling. General structuralism describes an item by its place in a relational organization. Inferentialism describes conceptual content through inferential role. Model-theoretic preservation describes invariance under transformations of structures. Transcendental arguments identify conditions of possibility for a practice, judgment, or experience.

The present theorem is narrower and more operational. It proves indispensability of a typed governance family under preservation of a fixed-domain reuse-relevant target. The proof does not claim that everything is relational, that concepts are inferential roles, or that a subject must presuppose what is denied. It says: once a same-domain distinction is treated as reuse-relevant, removing gate, identity, transport, quotient, redescription, boundary, or fail-closed roles prevents preservation of that target. The result is a fixed-domain continuation theorem, not a general metaphysical structuralism.

15 AMetric Boundary and No Pre-Governance Promotion

Lemma 15.1 (G13: Pre-admissible vacuity cannot select governance). *At the AMetric boundary, pre-admissible vacuity has no selector, ranking, parameter, metric, canonical representative, transport rule, default rule, exemption rule, or licensed operator by which it can generate governance on the fixed domain.*

Proof. A selector selects, a ranking orders, a parameter indexes, a metric measures, a canonical representative chooses, a transport rule licenses passage, a default rule privileges a candidate, and an exemption rule governs boundary crossing. Each is non-vacuous governance. If any is supplied, pre-admissible vacuity is no longer governance-vacuous. If none is supplied, no reuse-relevant promotion occurs. $\square$

Theorem 15.2 (G14: No governance-free lower generator). *No sub-admissible or pre-admissible governance-free layer can generate the governance bundle G_D for the same fixed domain without crossing the AMetric boundary by way of non-vacuous selector, witness, identity, transport, gate, boundary, or continuation structure.*

Proof. Let L be a proposed governance-free lower generator. If L supplies no selector, witness, identity carrier, transport rule, gate, boundary rule, or continuation discipline, it produces no reuse-relevant governance. If L supplies any of these, it is not governance-free. If those roles are supplied by a surrounding richer regime, the surrounding regime is non-vacuous or the domain has changed. Thus no governance-free lower generator exists for same-scope governance. $\square$

Corollary 15.3 (No default-governance rule). *A rule saying “in the absence of governance, governance is present by default” is not governance-free. It is a selector or default rule. It must either be independently admitted, in which case governance is non-vacuous, or fail to govern.*

16 Governance Equivalence under Redescription

Definition 16.1 (Governance equivalence). Two governance bundles G_D and G'_D are governance-equivalent, written $G_D \equiv_{\text{Gov}} G'_D$, iff they preserve identical admissibility-bearing continuation structure up to lawful redescription. Formally, $G_D \equiv_{\text{Gov}} G'_D$ iff they induce the same admissibility behavior, standing behavior, target-identity behavior, transport behavior, quotient behavior, continuation behavior, boundary behavior, and fail-closed behavior on the evaluated fragment E_D . Equivalence is therefore same-use invariant equivalence, not syntactic sameness of presentation. Invariant equivalence is always understood modulo lawful same-domain admissibility-preserving redescription.

Theorem 16.2 (G15: Uniqueness up to governance equivalence). *If two same-scope governance bundles preserve the same licensed use on D , then they are governance-equivalent:*

$$\text{SameScope}(G_D, G'_D) \wedge \text{SameUse}(G_D, G'_D) \rightarrow G_D \equiv_{\text{Gov}} G'_D.$$

Proof. If the two bundles preserve the same classifications, standing statuses, identities, transports, quotients, redescriptions, boundaries, and failures, they perform the same governance work by definition. If they differ only in notation, decomposition, or presentation, the difference is skin or bookkeeping. If they differ in licensed use, they are not SameUse. If they differ by adding hidden

selectors, witnesses, or carriers, they change governance and must either be included in the bundle or treated as domain change. Therefore same-scope same-use bundles are governance-equivalent. $\square$

Theorem 16.3 (G16: Constraint-induction uniqueness). *Any non-degenerate admissibility-bearing continuation regime induces an invariant-equivalent governance bundle in every same-domain realization preserving its continuation structure:*

$$\text{NonDeg}(D) \rightarrow \left(\text{Preserve}_{\mathcal{R}}(P_D) \rightarrow \exists G_D^{\mathcal{R}} (G_D^{\mathcal{R}} \equiv_{\text{Gov}} G_D) \right).$$

Proof. Assume $\text{NonDeg}(D)$, and let $\mathcal{R}$ preserve P_D . By the dependency-forcing ladder and the main theorem, $\mathcal{R}$ must contain invariant equivalents of identity, transport, quotient, gate, boundary, redescription, standing, and fail-closed treatment. Collect those equivalents into $G_D^{\mathcal{R}}$. Since they preserve the same classifications, standing statuses, target identities, transports, quotients, redescriptions, boundary statuses, and failure behavior required for P_D , $G_D^{\mathcal{R}} \equiv_{\text{Gov}} G_D$. Thus the governance bundle is structurally induced and unique up to governance equivalence. $\square$

Theorem 16.4 (G17: Constraint-induction closure). *Every non-degenerate admissibility-bearing continuation regime induces a unique governance-equivalence class of invariant constraint architecture:*

$$\text{NonDeg}(D) \rightarrow [G_D]_{\equiv_{\text{Gov}}}.$$

Equivalently, continuation preservation entails constraint architecture induction and equivalence-class uniqueness.

Proof. By G8, $\text{NonDeg}(D)$ structurally forces a non-vacuous governance bundle G_D . By G16, any same-domain realization preserving P_D induces a bundle $G_D^{\mathcal{R}}$ governance-equivalent to G_D . Therefore the induced object is not a preferred notation but the equivalence class $[G_D]_{\equiv_{\text{Gov}}}$: all same-domain preservation realizations fall into the same governance-equivalence class, and any realization outside that class fails to preserve the target phenomenon or changes domain. $\square$

Remark 16.5 (No preferred notation theorem). The theorem does not claim that the displayed tuple for G_D is the only possible notation. It claims that any same-domain notation must perform the same governance work or else fail to be same-use, same-scope governance.

17 Fixed-Domain Route Exhaustion

The main theorem can be attacked only by preserving $\text{NonDeg}(D)$ while attempting to remove G_D . The following factorization theorem records why those attempts are exhausted.

Theorem 17.1 (G18: Invariant-factorization exhaustion). *Every same-domain admissibility-bearing evaluability variation preserving P_D factorizes through one or more of the invariant loci*

$$\{\text{Id}, \text{Tr}, \text{Quot}, \text{Redesc}, \text{Adm}, \text{Stand}, \text{Fail}, \text{Bound}\}.$$

Any purported escape route outside this family is either scope change, bookkeeping, AMetric inertness, non-comparable presentation, or non-null governance under another name.

Proof. A same-domain variation preserving P_D must preserve the reuse-relevant distinction under continuation. Preservation of the same distinction factors through identity. Preservation across update factors through transport. Presentation variance factors through quotient and redescription. Licensed or blocked use factors through admissibility and standing. Boundary crossing, rejection, and non-evaluation factor through boundary typing. Failed standing under continuation factors through fail-closed treatment. These are precisely the listed invariant loci. A proposed route outside them either changes the target domain, changes no licensed use, remains AMetric inert, fails comparability, or supplies non-null governance that must be retyped into one of the listed loci. $\square$

Corollary 17.2 (Governance-vacuity route closure). *Every purported same-scope defense of non-degenerate but governance-vacuous evaluability falls into at least one of the following failure classes:*

- F1. **No-gate failure:** no licensed/unlicensed distinction remains.*
 - F2. **Domain-drift failure:** the proposal changes domain, evaluated fragment, predicate, boundary, or target.*
 - F3. **Empty-extension confusion:** object-level empty admission is mistaken for governance vacuity.*
 - F4. **Identity failure:** no stable target identity survives redescription.*
 - F5. **Transport failure:** no same target or status is carried through continuation.*
 - F6. **Quotient failure:** presentation-neutral comparison is undefined or load-erasing.*
 - F7. **Standing failure:** no commitment, continuation, or reuse can be standing-bearing.*
 - F8. **AMetric promotion failure:** the proposal relies on a selector, metric, parameter, or default that vacuity cannot supply.*
 - F9. **Skin-only authority failure:** presentation is asked to govern without anchor/tensor authority.*
 - F10. **Third-status failure:** a null-but-governing status is introduced between admitted and not admitted.*
 - F11. **Repair failure:** standing-null status is repaired into standing-positive status as the same act.*
 - F12. **Bookkeeping collapse:** the proposed difference changes no licensed use.*
- No further same-scope route remains under invariant-factorization exhaustion.*

18 Objection Matrix

Objection 1: The theorem is circular because governance is built into $\text{NonDeg}(D)$

Reply. The patched antecedent is intentionally non-degenerate rather than already-governed: $\text{NonDeg}(D)$ says only that at least one distinction is treated as reuse-relevant across lawful same-domain continuation. It does not include a gate, standing predicate, transport, quotient, redescription relation, boundary, or fail-closed repair rule. The dependency-forcing ladder derives those roles as preservation conditions.

Objection 2: This merely restates the null-regime theorem

Reply. The null-regime theorem proves that total AASC vacuity cannot occupy the admitted side of fixed-domain evaluability. The present theorem identifies the positive structural remainder after that closure: if fixed-domain reuse-relevant evaluability is in force, a non-vacuous governance bundle role-exists. Null closure blocks the endpoint; governance forcing identifies what the target phenomenon minimally requires.

Objection 3: Governance-existence is ontology in disguise

Reply. The theorem proves role-existence, not concrete existence. A role exists in this paper only when it is structurally required for preservation of the target phenomenon. The result does not introduce physical objects, events, fields, observers, spacetime, or unrestricted being.

Objection 4: An all-rejecting predicate admits nothing, so nothing exists

Reply. Correct: it admits no object-level candidate. But if the all-rejecting predicate functions as an admissibility-relevant classification, it still fixes the domain, the evaluated fragment, the predicate, the rejection/non-evaluation boundary, and the classification behavior. That is non-vacuous governance. Empty admission is not empty governance.

Objection 5: Why call this existence at all?

Reply. The term is indexed. $\exists_{\text{role}} G_D$ means that G_D is forced as role-content and cannot be removed without destroying, changing, or evacuating fixed-domain reuse-relevant evaluability. It does not mean that G_D is a concrete object.

Objection 6: The governance bundle is only notation

Reply. If the bundle changes no licensed use, it is notation or skin and is not the forced governance content. If it fixes admissibility, standing, identity, transport, quotient, redescription, boundary, or failure handling, it performs anchor/tensor work and is non-vacuous.

Objection 7: Could governance be external?

Reply. If external governance determines use-relevant status on D , it governs D and must be included in the fixed-domain role architecture or treated as a declared domain extension. If it determines nothing, it is inert. There is no outside-but-governing exception.

Objection 8: Could there be no domain at all?

Reply. Yes, but then $\text{NonDeg}(D)$ is denied. The theorem is conditional. It states that if non-degenerate fixed-domain evaluability is in force, governance cannot be vacuous. It does not prove that every domain is in force.

Objection 9: Does this prove an admitted occupant?

Reply. No. The all-rejecting evaluator has non-vacuous governance and empty admissibility extension. Occupant forcing is a later and stronger theorem.

Objection 10: Is this merely general structuralism?

Reply. No. General structuralism describes relational organization. This theorem proves role indispensability under fixed-domain preservation of a reuse-relevant distinction. It is not a broad metaphysics of structure, not inferentialism, and not a performative-transcendental argument.

Objection 11: What about a one-off distinction with no continuation?

Reply. Such a contrast is outside the main theorem's antecedent. The theorem is about $\text{NonDeg}(D)$, not every idle marking. A one-off distinction with no comparison, no continuation, no reuse, no exclusion, and no counterexample burden may be inert presentation. The theorem claims that once a distinction is treated as reuse-relevant across lawful same-domain continuation, invariant constraint architecture is forced.

19 Closure Ledger

Objection locus	Proposed escape	Closure mechanism
Circular antecedent	Governance hidden in the antecedent	$\text{NonDeg}(D)$ is continuation-targeted but not preloaded with gate, transport, quotient, or repair roles.
All-rejecting gate	No admitted objects, therefore vacuity	Empty admission is not empty governance.
No gate	Continuation without licensed/unlicensed distinction	The target phenomenon disappears.
External governance	Governance lies outside the domain	If it governs, it enters the role architecture or changes domain.
Pre-boundary source	Governance generated by null lower layer	AMetric boundary blocks selectors, parameters, metrics, and default promotion.
Pure notation	Governance as a symbol only	If it governs, anchor/tensor; if not, skin/bookkeeping.
Repair route	Failed standing becomes positive later	Fail-closed no same-act repair.
Third status	Null-but-governing status	Bivalence collapses to admitted/not-admitted or bookkeeping.

Alternative kernel	Weaker role package	Role removal destroys evaluability or reintroduces equivalent governance.
Structuralism objection	Merely relational relabeling	Fixed-domain continuation theorem with structural forcing and typed removal tests.
Concrete-overclaim	Governance means object existence	Role-existence is not concrete-existence.
Occupant demand	Non-vacuous governance must admit something	All-rejecting evaluator blocks the inference.

20 Canonical Results

20.1 Canonical one-line theorem

Any non-degenerate admissibility-bearing continuation regime forces non-vacuous invariant constraint architecture:

$$\text{NonDeg}(D) \rightarrow \text{CB}(G_D) \neq \text{Vac}.$$

20.2 Canonical structural-forcing formulation

$$P_D \Rightarrow_{\text{struct}} G_D.$$

Under $\text{NonDeg}(D)$, every same-domain realization preserving P_D contains an invariant-equivalent governance bundle, with invariant equivalence understood modulo lawful same-domain admissibility-preserving redescription.

20.3 Canonical role-existence formulation

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} G_D (\text{Gov}_D(G_D) \wedge \text{CB}(G_D) \neq \text{Vac}).$$

20.4 Canonical empty-extension formulation

An empty admissibility extension can be written or realized, but its admissibility-relevant classification is definable only relative to non-vacuous governance roles.

20.5 Canonical uniqueness formulation

Any non-degenerate admissibility-bearing continuation regime induces a governance bundle unique up to governance equivalence in every same-domain realization preserving its continuation structure:

$$\text{NonDeg}(D) \rightarrow \left(\text{Preserve}_{\mathcal{R}}(P_D) \rightarrow \exists G_D^{\mathcal{R}} (G_D^{\mathcal{R}} \equiv_{\text{Gov}} G_D) \right).$$

20.6 Canonical constraint-induction closure formulation

Every non-degenerate admissibility-bearing continuation regime induces a unique governance-equivalence class of invariant constraint architecture:

$$\text{NonDeg}(D) \rightarrow [G_D]_{\equiv_{\text{Gov}}}.$$

20.7 Canonical anti-overclaim formulation

The theorem proves role-existence of invariant constraint architecture. It does not prove admitted occupants, concrete instantiation, physical ontology, or unrestricted being.

21 Short Proof Form

1. $\text{NonDeg}(D)$ means that at least one distinction is treated as reuse-relevant across lawful same-domain continuation.
2. Reuse-relevant continuation requires persistence criteria.
3. Persistence criteria require identity fixation.
4. Identity across update requires transport.
5. Transport under presentation variance requires quotient and redescription discipline.
6. Quotient-respecting continuation requires a gate distinguishing licensed from unlicensed standing-bearing use.
7. Gate authority requires domain identity, predicate identity, evaluated-fragment identity, and boundary typing.
8. Standing-bearing classification requires standing, identity, transport, quotient consistency, and redescription stability on the positive side.
9. Failed standing must be fail-closed for the same evaluated act.
10. These roles form $G_D = \langle \text{Gate}, \text{Adm}, \text{Stand}, \text{Id}, \text{Tr}, \text{Quot}, \text{Redesc}, \text{Bound}, \text{Fail} \rangle$.
11. Removing any role without equivalent replacement destroys preservation of P_D .
12. Therefore $P_D \Rightarrow_{\text{struct}} G_D$, and $\text{CB}(G_D) \neq \text{Vac}$.
13. Any same-domain realization preserving P_D induces a governance-equivalent bundle, so $\text{NonDeg}(D) \rightarrow [G_D]_{\equiv_{\text{Gov}}}$.
14. Empty object-level admission does not imply empty governance, because extension classification itself requires domain, predicate, boundary, and classification roles.
15. The result is a positive structural forcing theorem, not an ontology or occupant-existence theorem.

22 Paper III Boundary

The present paper ends at non-vacuous governance. It intentionally does not prove occupant forcing. The next theorem in the sequence would require an additional premise stronger than $\text{NonDeg}(D)$, because the all-rejecting evaluator shows that governance can be non-vacuous while admission extension is empty.

A candidate Paper III target would be:

$$\text{NonDeg}(D) \wedge P_D \rightarrow \exists x \in E_D \text{ Adm}_D(x) = 1,$$

where P_D must block the all-rejecting evaluator without smuggling in the desired occupant. Possible versions of P_D include non-degenerate commitment, unavoidable standing-bearing participation, constructional success, or unrestricted evaluability. Those conditions are not assumed here.

The final and still stronger target would be concrete instantiation:

$$\exists x \text{ Concrete}(x),$$

which would require a further bridge from admitted structural occupants to concrete ontology. Nothing in the present theorem supplies that bridge.

23 Conclusion

The null-regime theorem showed that total AASC vacuity cannot occupy the admitted side of fixed-domain evaluability. The present paper identifies the positive structural remainder after that closure. Non-degenerate admissibility-bearing continuation cannot be governance-vacuous. If a same-domain regime treats a distinction as reuse-relevant across lawful continuation, it must preserve persistence, identity, transport, quotient consistency, redescription stability, gate authority, boundary typing, standing eligibility, and fail-closed standing failure. These roles form a typed governance bundle.

The result is:

$$\text{NonDeg}(D) \rightarrow \exists_{\text{role}} G_D (\text{Gov}_D(G_D) \wedge \text{CB}(G_D) \neq \text{Vac}),$$

or equivalently:

$$P_D \Rightarrow_{\text{struct}} G_D.$$

This is the positive structural successor to the null-regime closure theorem. Its endpoint is stronger than bare governance non-vacuity: admissibility-bearing continuation induces invariant constraint architecture uniquely up to lawful redescription and governance equivalence. In compact form, continuation preservation entails constraint architecture induction and equivalence-class uniqueness. It does not prove object-level admission, concrete instantiation, physical spacetime, observers, fields, particles, events, or unrestricted ontology. The next paper in the arc may ask whether additional assumptions can force an admitted occupant. The present result deliberately stops at the unique governance-equivalence class induced by non-degenerate continuation.

24 Counterexample-Burden Appendix

A successful counterexample to the main theorem would have to satisfy all of the following conditions simultaneously:

1. preserve $\text{NonDeg}(D)$, meaning that at least one distinction remains reuse-relevant across lawful same-domain continuation;
2. remove every non-vacuous governance role;
3. retain a same fixed domain and same evaluated fragment;
4. distinguish admissibility-relevant rejection from non-evaluation without a gate;
5. define an admissibility extension without predicate identity or boundary typing;
6. preserve target identity without identity structure;
7. preserve continuation without transport;
8. preserve presentation-neutral comparison without quotient structure;
9. preserve standing-bearing reuse without standing;
10. cross the AMetric boundary without selector, witness, parameter, metric, default, or gate;
11. govern by skin-only presentation without anchor or tensor authority;
12. avoid bivalence while remaining same-scope; and
13. repair standing failure while remaining the same evaluated act.

These burdens cannot be jointly satisfied. The first condition keeps the target phenomenon in place. The second removes the roles required by that phenomenon. Conditions 3–13 are exactly the fixed-domain loci at which the contradiction becomes explicit. Therefore no same-scope governance-vacuous evaluator exists.

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